Math 534 H - Spring 2026 - Lecture 12

Wave equation <u>conserves energy</u>.

1. We are going to assume that the IC ϕ and ψ vanish outside an interval $[-R, R]$

This implies $u(x,t)$ (& hence $u_x(x,t)$ and $u_t(x,t)$) vanish for $|x| > R + ct$.

ρ : density of the string $\rho > 0$

T : magnitude of the tension vector (constant) $T > 0$

(ρ, T indep of x, t) and $\boxed{c^2 = \frac{T}{\rho}}$

c = speed of wave

Eq. for the string is $\rho u_{tt} = T u_{xx}$ $x \in \mathbb{R}$.

$$u_{tt} - \frac{T}{\rho} u_{xx} = 0 = u_{tt} - c^2 u_{xx}$$

<u>Kinetic Energy</u> $KE = \frac{1}{2}\rho \int_{-\infty}^{\infty} u_t^2 \, dx$

$$PE = \frac{1}{2} T \int_{-\infty}^{\infty} u_x^2 \, dx$$

$u_t = u_t(x,t)$

$u_x = u_x(x,t)$

$E = KE + PE$ energy functional for u

$$E[u](t) = \frac{1}{2}\rho \int_{-\infty}^{\infty} u_t^2(x,t)\,dx + \frac{1}{2} T \int_{-\infty}^{\infty} u_x(x,t)\,dx$$

Conservation of Energy means

$$E[u](t) = E[u](0) \longrightarrow \text{constant}. \Longleftrightarrow$$

$$\frac{dE}{dt} = 0 = \frac{dKE}{dt} + \frac{dPE}{dt}$$

Let's prove this. WLOG, $\rho = 1 = T$, $c^2 = 1$

$$\frac{dKE}{dt} = \frac{1}{2}\int_{-\infty}^{\infty} \frac{d}{dt}(u_t^2)\,dx = \frac{1}{2}\cdot 2\int_{-\infty}^{\infty} u_t\, u_{tt}\,dx$$

But $u_{tt} - u_{xx} = 0$ $\quad u_{tt} = u_{xx}$

$$\frac{dKE}{dt} = \int_{-\infty}^{\infty} u_t\, u_{xx} \overset{IBP}{=} \underbrace{u_t\, u_x \Big|_{-\infty}^{\infty}}_{0} - \int_{-\infty}^{\infty} u_{tx}\, u_x\,dx$$

$$\rightarrow \quad \frac{dKE}{dt} = -\int_{-\infty}^{\infty} \underbrace{u_{tx}\, u_x}_{= \frac{1}{2}\frac{d}{dt}\left(u_x^2\right)} \, dx$$

$$\frac{dKE}{dt} = -\frac{1}{2}\frac{d}{dt}\int_{-\infty}^{\infty} u_x^2 \, dx \qquad (1)$$

But $$\frac{dPE}{dt} = \frac{1}{2}\frac{d}{dt}\int_{-\infty}^{\infty} u_x^2 \, dx \qquad (2)$$

Then $$\frac{dE}{dt} = \underbrace{\frac{dKE}{dt}}_{(1)} + \underbrace{\frac{dPE}{dt}}_{(2)} = 0 \quad !$$

as desired.

Another Examples

1. Damped string equation. This arises from string/wave problem when significant - say - air resistance is at play. r = resistance factor $\underline{r > 0}$.

$$\boxed{u_{tt} - c^2 u_{xx} + \underline{r u_t} = 0}$$

For this equation $E = KE + PE$ as before.

$$E[u](t) = \frac{1}{2}\rho \int_{-\infty}^{\infty} u_t^2 \, dx + \frac{1}{2} T \int_{-\infty}^{\infty} u_x^2 \, dx$$

Repeat same as before but now write

$$u_{tt} = c^2 u_{xx} - r u_t$$ Extra!

So now you'll get

$$\frac{d\,KE}{dt} = \cancel{-\frac{1}{2} T \frac{d}{dt} \int_{-\infty}^{\infty} u_x^2 \, dx} - \rho r \int_{-\infty}^{\infty} u_t^2 \, dx$$

$$\frac{dPE}{dt} = \cancel{\frac{1}{2} T \frac{d}{dt} \int_{-\infty}^{\infty} u_x^2 \, dx}$$

survives!

$$\frac{dE}{dt} = \frac{d\,KE}{dt} + \frac{dPE}{dt} = \underbrace{-\rho r \int_{-\infty}^{\infty} u_t^2 \, dx}_{\text{negative!}} \leq 0$$

$$\underbrace{E[u](t)}_{\text{decreasing}} \leq E[u](0)$$

2 Suppose now in your string problem there is a transverse elastic force $\Rightarrow$ we have an extra term proportional to the displacement u (like in a coiled string)

Then the equation is

$$\boxed{\begin{gathered} u_{tt} - c^2 u_{xx} + k u = 0 \\ k > 0 \end{gathered}}$$

Try to compute $\frac{dE}{dt}$ where E is the same as before. you end up with an extra term that looks like $-\rho k \int_{-\infty}^{\infty} u_t u \, dx$ which has <u>no sign</u>!

Do we need to modify the energy! How do we find?

$$0 = u_{tt} - c^2 u_{xx} + k u$$

$$0 = \left(u_{tt} - c^2 u_{xx} + k u \right) u_t$$

$$0 = \underbrace{u_{tt} \cdot u_t}_{\frac{1}{2}\frac{d}{dt}(u_t)^2} \underbrace{- c^2 u_{xx} u_t}_{} + \underbrace{k u u_t}_{\frac{1}{2} k \frac{d}{dt}(u^2)}.$$

Next integrate both sides w.r.t x and do IBP in the second term on the r.h.s as before.

For simplicity here assume $c^2 = 1$ $k = 1$ $T = 1$ $\rho = 1$

Then we'll get

$$0 = \frac{1}{2}\frac{d}{dt}\left[\underbrace{\int \left(u_t^2 + u_x^2 + u^2\right) dx}_{\tilde{E}[u](t)}\right]$$

new energy!

This $\tilde{E}[u]$ is conserved in time.

We'll do this last part in detail next time...